```
import pandas as pd
import matplotlib.pyplot as plt

ds = pd.read_csv("wan_dataset.csv")
ds['timestamp'] = pd.to_datetime(ds['timestamp'])
print(ds.head())
```

```
   timestamp link_id source_device destination_device source_interface  \
0 1970-01-01  link_1        host_1           switch_1      interface_1
1 1970-01-01  link_2        host_2           switch_3      interface_2
2 1970-01-01  link_3        host_3           switch_1      interface_3
3 1970-01-01  link_4        host_4           switch_3      interface_4
4 1970-01-01  link_5        host_5           switch_1      interface_5

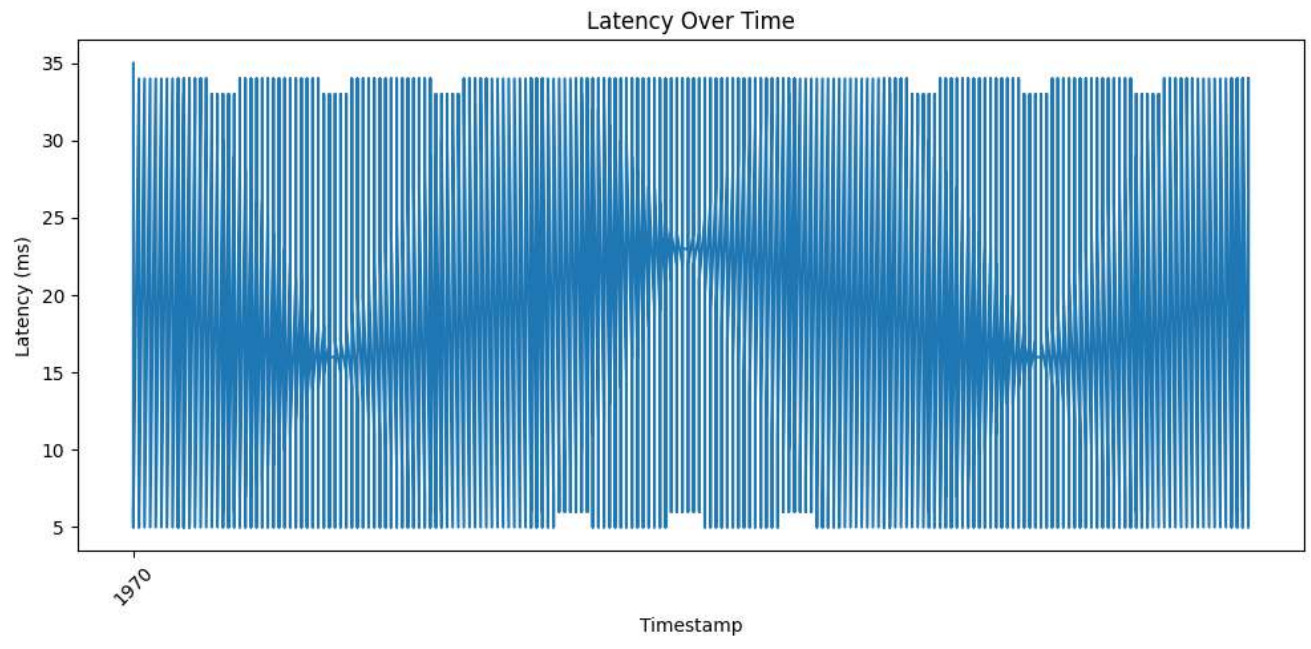
  destination_interface  link_utilization  latency_ms  packet_loss  \
0           interface_1                50          35          0.0
1           interface_4                83          28          1.2
2           interface_7                86          13          0.4
3          interface_10                55           5          1.6
4           interface_1                19          10          0.8

   throughput_mbps  qos_priority  reachability   traffic_type  anomaly_flag
0              200             1             0           http             1
1              452             2             1          video             0
2              472             3             1           voip             0
3              242             4             1  file_transfer             0
4              -27             5             1           http             0
```

1. Line Plot

This plot shows how latency changes over time.

```
plt.figure(figsize=(10, 5))
plt.plot(ds['timestamp'], ds['latency_ms'])
plt.title("Latency Over Time")
plt.xlabel("Timestamp")
plt.ylabel("Latency (ms)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

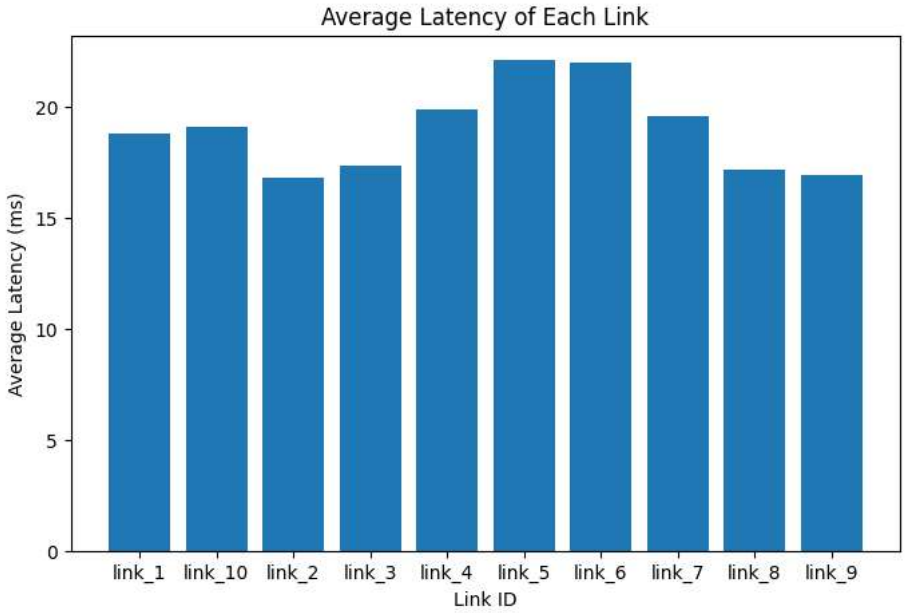


2. Bar Plot

This plot compares the average latency of different links.

```
avg_latency = ds.groupby('link_id')['latency_ms'].mean()

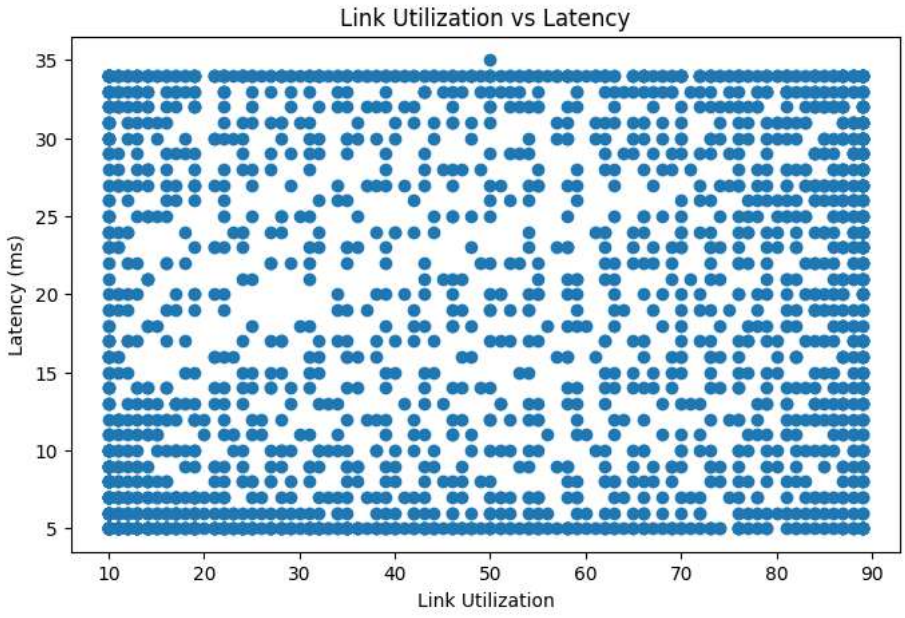
plt.figure(figsize=(8, 5))
plt.bar(avg_latency.index.astype(str), avg_latency.values)
plt.title("Average Latency of Each Link")
plt.xlabel("Link ID")
plt.ylabel("Average Latency (ms)")
plt.show()
```



3. Scatter Plot

This plot shows the relationship between link utilization and latency.

```
plt.figure(figsize=(8, 5))
plt.scatter(ds['link_utilization'], ds['latency_ms'])
plt.title("Link Utilization vs Latency")
plt.xlabel("Link Utilization")
plt.ylabel("Latency (ms)")
plt.show()
```

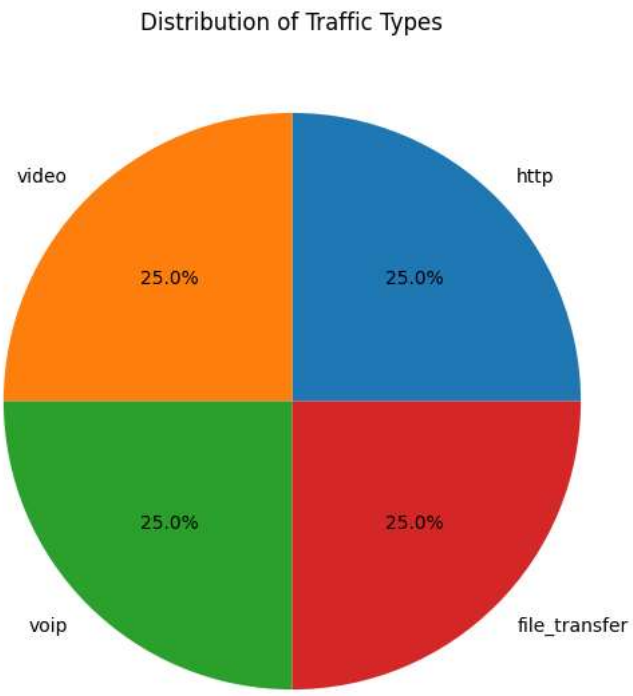


4. Pie Plot

This plot shows the proportion of each traffic type.

```
traffic_count = ds['traffic_type'].value_counts()

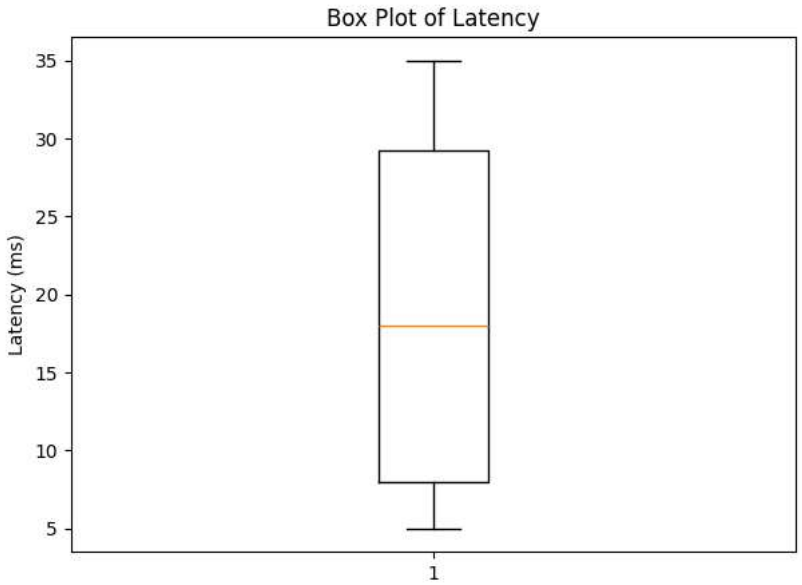
plt.figure(figsize=(7, 7))
plt.pie(traffic_count, labels=traffic_count.index, autopct='%1.1f%%')
plt.title("Distribution of Traffic Types")
plt.show()
```



5. Box Plot

This plot displays the distribution of latency values.

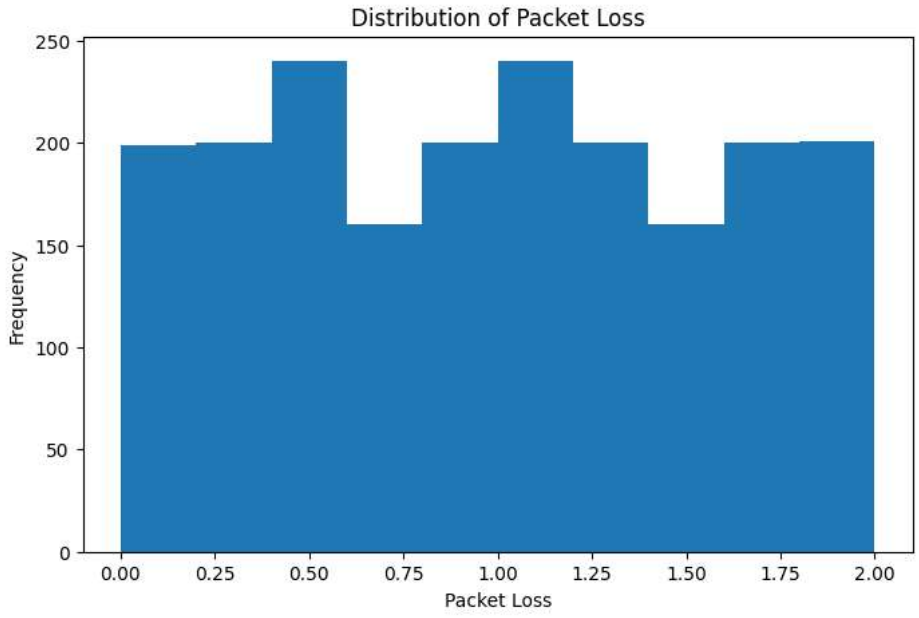
```
plt.figure(figsize=(7, 5))
plt.boxplot(ds['latency_ms'].dropna())
plt.title("Box Plot of Latency")
plt.ylabel("Latency (ms)")
plt.show()
```



6. Histogram

This plot shows the frequency distribution of packet loss.

```
plt.figure(figsize=(8, 5))
plt.hist(ds['packet_loss'].dropna(), bins=10)
plt.title("Distribution of Packet Loss")
plt.xlabel("Packet Loss")
plt.ylabel("Frequency")
plt.show()
```



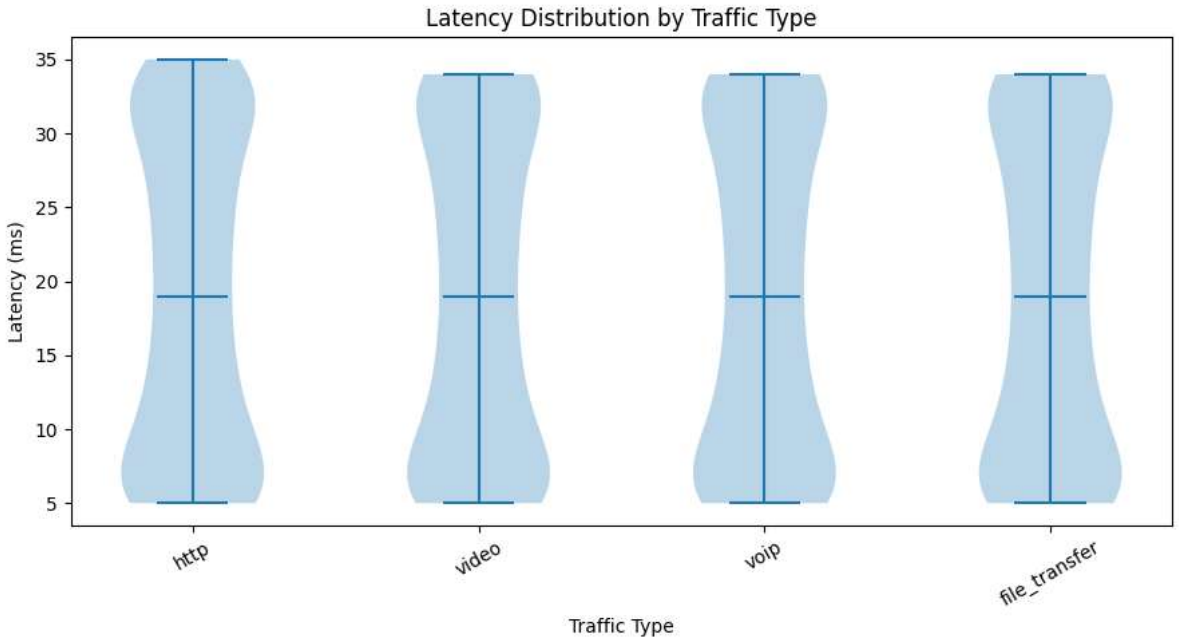
7. Violin Plot

This plot shows the distribution of latency values for each traffic type.

```
traffic_types = ds['traffic_type'].dropna().unique()
latency_groups = []

for t in traffic_types:
    latency_groups.append(ds[ds['traffic_type'] == t]['latency_ms'].dropna())

plt.figure(figsize=(9, 5))
plt.violinplot(latency_groups, showmeans=True)
plt.title("Latency Distribution by Traffic Type")
plt.xlabel("Traffic Type")
plt.ylabel("Latency (ms)")
plt.xticks(range(1, len(traffic_types) + 1), traffic_types, rotation=30)
plt.tight_layout()
plt.show()
```



8. Heatmap

This plot uses colours to show the correlation between numerical WAN attributes.

```
correlation = ds[['link_utilization', 'latency_ms', 'packet_loss', 'throughput_mbps']].corr()

plt.figure(figsize=(7, 5))
plt.imshow(correlation, cmap='viridis')
```